You said:



Assignment - Pyth...
Document

Hello Shanmuga,

Congratulations! Your profile has been shortlisted for the Python/Django Developer Intern opportunity.

Please complete and submit the assignment within 24 hours of receiving this email.

Assignment Deliverables:

- GitHub Repository Link
- README.md
- AI_USAGE.md (including AI tools used, prompts, accepted/rejected outputs, mistakes identified, and verification steps)
- Exported chat transcript attached and included in the repository
- Test Cases
- Chat Transcript File
- Test Run Output (either a GitHub Actions link or a TEST_OUTPUT.md file containing terminal output)

Please share all the required deliverables in a single ZIP file for ease of review.

Job Requirements Confirmation:

Additionally, please confirm your alignment with the following internship requirements:

- Location: Are you open to relocating to Koregaon Park, Pune, for this on-site role?
- Availability: What is your earliest availability to join?
- Stipend: Are you comfortable with a monthly stipend of INR 13,000 - 15,100?

We look forward to your submission.

--

Thanks and Regards,
Khushi Walje,
Human Resource Department,
Tradexa Technologies Pvt Ltd. Hey I got shortlisted for an intern in Pune. But zthe next shortlisting process needs this project to be done and submit it within 24 hrs so help me with it and actually before going to code gimme all the possible solutions can be given and finalize the one that'll be the best suit for this project.

ChatGPT said:

You said:

Finalize the box selection assignment design. Packing approach: 3D packing heuristic supporting multiple products; interface: Django REST API plus admin panel. Now guide me through implementing the project step by step, with complete files, tests, and honest assignment documentation.

Analyzed

ChatGPT said:

‹/›

You said:

Hey the step 1 and 2 works and the step 3 throws some errors should I must create a file named manage.py in the folder

ChatGPT said:

You said:

The dir has only .venv and the 0 as length for requirements.txt

Analyzed

ChatGPT said:

‹/›

You said:

should i go and repeat the same steps to create a .venv.. reply in short

ChatGPT said:

You said:

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>python manage.py createsuperuser
Username (leave blank to use 'shan'): shan
Email address: shanmugasundaramdm@gmail.com
Password:
Password (again):
Superuser created successfully.

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
October 19, 2026 - 18:17:43
Django version 5.2.18, using settings 'config.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/
Not Found: /
[19/Oct/2026 18:18:01] "GET / HTTP/1.1" 404 2304

ChatGPT said:

You said:

So gimme the lists of some sample products and the shipping boxes

ChatGPT said:

You said:

Pasted text(3).txt
Document

i got this error when try to click the save and add another -

ChatGPT said:

You said:

{
"items": [
{"product_id": 1, "quantity": 1},
{"product_id": 3, "quantity": 2}
]
} I have added this so what to do next also where can see the result adding this json in the same /recommend place or somewhere else and what to do next

ChatGPT said:

You said:

HTTP 200 OK Allow: POST, OPTIONS **Content-Type:** application/json **Vary:** Accept { "found": true, "box": { "id": 2, "name": "Medium Box", "cost_inr": "50.00", "max_weight_kg": "5.000" }, "total_order_weight_kg": "1.200", "packing_method": "3d_corner_point_heuristic", "placements": [{ "item_key": "3-1", "position_cm": { "x": 0.0, "y": 0.0, "z": 0.0 }, "oriented_dimensions_cm": { "length": 8.0, "width": 18.0, "height": 20.0 } }, { "item_key": "3-2", "position_cm": { "x": 8.0, "y": 0.0, "z": 0.0 }, "oriented_dimensions_cm": { "length": 8.0, "width": 18.0, "height": 20.0 } }, { "item_key": "1-1", "position_cm": { "x": 16.0, "y": 0.0, "z": 0.0 }, "oriented_dimensions_cm": { "length": 2.0,

"width": 8.0, "height": 15.0 } }], "note": "Lowest-cost box accepted by the packing heuristic; not a proof of globally optimal packing.

ChatGPT said:

You said:

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>python manage.py test
Found 11 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
.....C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\fields.py:992: UserWarning: min_value should be an integer or Decimal instance.
  warnings.warn("min_value should be an integer or Decimal instance.")
.....

Ran 11 tests in 0.032s

OK
Destroying test database for alias 'default'...

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>python manage.py check
System check identified no issues (0 silenced).

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system> so can we do the edge test cases gimme the required details - i got an error raised for the first 2 test cases and for the 3rd and 4th one there should be a new product and new weight that is over that the box limits should be given right

ChatGPT said:

You said:

Box_Selection_Test...
Document

Here is the all four test cases that has been done so what do we do next

ChatGPT said:

You said:

So in the test cases.md I should give the topics of the test cases that I have ran so for the topics of 5 test cases ha and in the test ouput should I give the results of them the same one I shared here with you. and the test 4 fix response - **HTTP 200 OK Allow:** POST, OPTIONS **Content-Type:** application/json **Vary:** Accept { "found": false, "message": "No available box passed the weight check and packing heuristic.", "total_order_weight_kg": "26.000", "rejected_boxes": [{ "box": "Small Box", "reason": "order_exceeds_weight_capacity" }, { "box": "Medium Box", "reason": "order_exceeds_weight_capacity" }, { "box": "Large Box", "reason": "order_exceeds_weight_capacity" }, { "box": "Heavy Duty Box", "reason": "order_exceeds_weight_capacity" }, { "box": "Extra Large Box", "reason": "order_exceeds_weight_capacity" }], "packing_method": "3d_corner_point_heuristic", "limitation": "A heuristic failure does not prove that no feasible physical arrangement exists."

ChatGPT said:

You said:

Test Cases

Automated tests are in `inventory/tests.py`.

ID	Scenario	Expected behavior
TC-01	Single product fits the cheapest box	Recommend the cheapest qualifying box
TC-02	Product requires axis-aligned rotation	Packing succeeds if the rotated dimensions fit
TC-03	Product is too large for candidate box	Candidate is rejected
TC-04	Order exceeds a box's weight limit	Box is rejected and a later candidate may be selected
TC-05	Multiple items fit without overlap	Return a placement for every item
TC-06	No available box can fit item	Return `found: false`
TC-07	Empty order	Return HTTP 400
TC-08	Unknown product ID	Return HTTP 400
TC-09	Duplicate product entries	Return HTTP 400 and request one entry with quantity
TC-10	Quantity is zero	Return HTTP 400
TC-11	Product and box list endpoints	Return HTTP 200
TC-12	Equal cost	Lowest database ID is considered first

Run with `python manage.py test`. Add additional tests after you manually review edge cases. what about this should I clear it or have it in the test cases .md

ChatGPT said:

You said:

so can i erase or keep the 12 tc

ChatGPT said:

You said:

these are the contents in test output.md can i clear it

Worked for 9s

ChatGPT said:

You said:

Environment:
- OS:
- Python version (`python --version`):
- Django version (`python -m django --version`): should we need this in the test_output.md and here are the contents in test_cases.md - **# Test Cases**

Automated tests are in `inventory/tests.py`.

| ID | Scenario | Expected behavior |

TC-01	Single product fits the cheapest box	Recommend the cheapest qualifying box
TC-02	Product requires axis-aligned rotation	Packing succeeds if the rotated dimensions fit
TC-03	Product is too large for candidate box	Candidate is rejected
TC-04	Order exceeds a box's weight limit	Box is rejected and a later candidate may be selected
TC-05	Multiple items fit without overlap	Return a placement for every item
TC-06	No available box can fit item	Return `found: false`
TC-07	Empty order	Return HTTP 400
TC-08	Unknown product ID	Return HTTP 400
TC-09	Duplicate product entries	Return HTTP 400 and request one entry with quantity
TC-10	Quantity is zero	Return HTTP 400
TC-11	Product and box list endpoints	Return HTTP 200

Run with `python manage.py test`. Add additional tests after you manually review edge cases.

Test Cases

1. Valid Order

- Input: One smartphone and two headphones.
- Expected: Recommend the lowest-cost box the packing heuristic can fit all items into.

- Actual: Medium Box recommended at ₹50.
- Status: Passed.

2. Zero Quantity

- Input: Product ID 1 with quantity 0.
- Expected: HTTP 400 validation error.
- Actual: HTTP 400; quantity must be at least 1.
- Status: Passed.

3. Unknown Product ID

- Input: Product ID 10, which does not exist.
- Expected: HTTP 400 validation error.
- Actual: HTTP 400; unknown product ID reported.
- Status: Passed.

4. Oversized Product

- Input: Oversized Monitor (100 × 80 × 60 cm).
- Expected: No box recommendation if the packing heuristic cannot fit the product.
- Actual: HTTP 200; `found: false`.
- Status: Passed, with heuristic limitation noted.

5. Order Exceeds Weight Capacity

- Input: Heavy Test Item weighing 26 kg.
- Expected: No box recommendation because every box has a maximum weight below 26 kg.
- Actual: HTTP 200; `found: false`. All five boxes were rejected for exceeding weight capacity.
- Status: Passed.
and here are the test_output.md - **# Test Run Output**

Paste the actual terminal output from your own run of:

- python manage.py test

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py test
Found 11 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
.....C:\Users\Shan\Downloads\box_selection_system_starter\box_sele
ction_system\.venv\Lib\site-packages\rest_framework\fields.py:992:
UserWarning: min_value should be an integer or Decimal instance.
  warnings.warn("min_value should be an integer or Decimal
instance.")
......
----------------------------------------------------------------------
Ran 11 tests in 0.034s

OK
Destroying test database for alias 'default'...
```

- python manage.py check

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py check
System check identified no issues (0 silenced).
```

Environment:
- OS:
- Python version (`python --version`):
- Django version (`python -m django --version`):

Actual terminal output:

`Summarization` : In below are the test cases, that has been tested by giving the correct valid order values, quantity as 0, unknown pr0duct id that is not mentioned or added in the product (product_id = 15), product that is too large for every box to carry by, the order exceeds every boxes weight limits here by these are the test cases conducted on this project.

1. Valid Order

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "found": true,
    "box": {
        "id": 2,
        "name": "Medium Box",
        "cost_inr": "50.00",
        "max_weight_kg": "5.000"
    },
    "total_order_weight_kg": "1.200",
    "packing_method": "3d_corner_point_heuristic",
    "placements": [
        {
            "item_key": "3-1",
            "position_cm": {
                "x": 0.0,
                "y": 0.0,
                "z": 0.0
            },
            "oriented_dimensions_cm": {
                "length": 8.0,
                "width": 18.0,
                "height": 20.0
            }
        },
        {
            "item_key": "3-2",
            "position_cm": {
                "x": 8.0,
                "y": 0.0,
                "z": 0.0
            },
            "oriented_dimensions_cm": {
                "length": 8.0,
                "width": 18.0,
```

```
            "height": 20.0
          }
        },
        {
          "item_key": "1-1",
          "position_cm": {
            "x": 16.0,
            "y": 0.0,
            "z": 0.0
          },
          "oriented_dimensions_cm": {
            "length": 2.0,
            "width": 8.0,
            "height": 15.0
          }
        }
    ],
    "note": "Lowest-cost box accepted by the packing heuristic; not a proof of globally optimal packing."
```


2. QUANTITY = 0

HTTP 400 Bad Request
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "error": "Invalid request",
    "details": {
        "items": [
            {
                "quantity": [
                    "Ensure this value is greater than or equal to 1."
                ]
            }
        ]
    }
}
```


3. UNKNOWN PRODUCT ID = 15

HTTP 400 Bad Request
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "error": "Invalid request",
    "details": {
        "items": [
            "Unknown product_id value(s): [15]"
        ]
    }
}
```


4. PRODUCT TOO LARGE FOR EVERY BOX - OVERSIZED MONITOR

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
    "found": false,
    "message": "No available box passed the weight check and packing heuristic.",
    "total_order_weight_kg": "3.000",
    "rejected_boxes": [
        {
            "box": "Small Box",
            "reason": "order_exceeds_weight_capacity"
        },
        {
            "box": "Medium Box",
            "reason": "heuristic_could_not_pack_all_items"
        },
        {
            "box": "Large Box",
            "reason": "heuristic_could_not_pack_all_items"
        },
        {
```

```
            "box": "Heavy Duty Box",
            "reason": "heuristic_could_not_pack_all_items"
        },
        {
            "box": "Extra Large Box",
            "reason": "heuristic_could_not_pack_all_items"
        }
    ],
    "packing_method": "3d_corner_point_heuristic",
    "limitation": "A heuristic failure does not prove that no feasible
physical arrangement exists.
```


5. ORDER EXCEEDS EVERY BOX'S WEIGHT LIMIT - QUANTITY - 101

HTTP 400 Bad Request
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "error": "Invalid request",
    "details": {
        "items": [
            {
                "quantity": [
                    "Ensure this value is less than or equal to 100."
                ]
            }
        ]
    }
}
```


Fixing Test 5: Testing the actual weight limit

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept

"found": false,
"message": "No available box passed the weight check and packing heuristic.",
"total_order_weight_kg": "26.000",
"rejected_boxes": [
{
"box": "Small Box",
"reason": "order_exceeds_weight_capacity"
},
{
"box": "Medium Box",
"reason": "order_exceeds_weight_capacity"
},
{
"box": "Large Box",
"reason": "order_exceeds_weight_capacity"
},
{
"box": "Heavy Duty Box",
"reason": "order_exceeds_weight_capacity"
},
{
"box": "Extra Large Box",
"reason": "order_exceeds_weight_capacity"
}
],
"packing_method": "3d_corner_point_heuristic",
"limitation": "A heuristic failure does not prove that no feasible physical arrangement exists."

NOTE: Total weight = (0.2 * 101 = 20.2 kg).

ChatGPT said:

You said:

Test Run Output

Environment

- OS: Windows 11
- Python version: Python 3.11.6
- Django version: 5.2.18

Automated Test Results

Command: `python manage.py test`

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py test
Found 11 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
.....C:\Users\Shan\Downloads\box_selection_system_starter\box_sele
ction_system\.venv\Lib\site-packages\rest_framework\fields.py:992:
UserWarning: min_value should be an integer or Decimal instance.
  warnings.warn("min_value should be an integer or Decimal
instance.")
......
----------------------------------------------------------------------
Ran 11 tests in 0.034s

OK
Destroying test database for alias 'default'...
```

Django System Check

Command: `python manage.py check`

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py check
System check identified no issues (0 silenced).
```

Manual API Test Results

`Summarization` : In below are the test cases, that has been tested by giving the correct valid order values, quantity as 0, unknown pr0duct id that is not mentioned or added in the product

(product_id = 15), product that is too large for every box to carry by, the order exceeds every boxes weight limits here by these are the test cases conducted on this project.

Actual terminal output:

1. Valid Order

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
  "found": true,
  "box": {
    "id": 2,
    "name": "Medium Box",
    "cost_inr": "50.00",
    "max_weight_kg": "5.000"
  },
  "total_order_weight_kg": "1.200",
  "packing_method": "3d_corner_point_heuristic",
  "placements": [
    {
      "item_key": "3-1",
      "position_cm": {
        "x": 0.0,
        "y": 0.0,
        "z": 0.0
      },
      "oriented_dimensions_cm": {
        "length": 8.0,
        "width": 18.0,
        "height": 20.0
      }
    },
    {
      "item_key": "3-2",
      "position_cm": {
        "x": 8.0,
```

```
            "y": 0.0,
            "z": 0.0
        },
        "oriented_dimensions_cm": {
            "length": 8.0,
            "width": 18.0,
            "height": 20.0
        }
    },
    {
        "item_key": "1-1",
        "position_cm": {
            "x": 16.0,
            "y": 0.0,
            "z": 0.0
        },
        "oriented_dimensions_cm": {
            "length": 2.0,
            "width": 8.0,
            "height": 15.0
        }
    }
],
"note": "Lowest-cost box accepted by the packing heuristic; not a proof of globally optimal packing."
```


2. QUANTITY = 0

HTTP 400 Bad Request
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "error": "Invalid request",
    "details": {
        "items": [
            {
                "quantity": [
                    "Ensure this value is greater than or equal to 1."
                ]
```

```
            }
        ]
    }
}
```


3. UNKNOWN PRODUCT ID = 15

HTTP 400 Bad Request
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "error": "Invalid request",
    "details": {
        "items": [
            "Unknown product_id value(s): [15]"
        ]
    }
}
```


4. PRODUCT TOO LARGE FOR EVERY BOX - OVERSIZED MONITOR

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
    "found": false,
    "message": "No available box passed the weight check and packing heuristic.",
    "total_order_weight_kg": "3.000",
    "rejected_boxes": [
        {
            "box": "Small Box",
            "reason": "order_exceeds_weight_capacity"
        },
        {
            "box": "Medium Box",
            "reason": "heuristic_could_not_pack_all_items"
```

```
        },
        {
            "box": "Large Box",
            "reason": "heuristic_could_not_pack_all_items"
        },
        {
            "box": "Heavy Duty Box",
            "reason": "heuristic_could_not_pack_all_items"
        },
        {
            "box": "Extra Large Box",
            "reason": "heuristic_could_not_pack_all_items"
        }
    ],
    "packing_method": "3d_corner_point_heuristic",
    "limitation": "A heuristic failure does not prove that no feasible
physical arrangement exists.
```


5. ORDER EXCEEDS EVERY BOX'S WEIGHT LIMIT - QUANTITY - 101

HTTP 400 Bad Request
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "error": "Invalid request",
    "details": {
        "items": [
            {
                "quantity": [
                    "Ensure this value is less than or equal to 100."
                ]
            }
        ]
    }
}
```


additional validation check:

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
    "found": false,
    "message": "No available box passed the weight check and packing heuristic.",
    "total_order_weight_kg": "26.000",
    "rejected_boxes": [
        {
            "box": "Small Box",
            "reason": "order_exceeds_weight_capacity"
        },
        {
            "box": "Medium Box",
            "reason": "order_exceeds_weight_capacity"
        },
        {
            "box": "Large Box",
            "reason": "order_exceeds_weight_capacity"
        },
        {
            "box": "Heavy Duty Box",
            "reason": "order_exceeds_weight_capacity"
        },
        {
            "box": "Extra Large Box",
            "reason": "order_exceeds_weight_capacity"
        }
    ],
    "packing_method": "3d_corner_point_heuristic",
    "limitation": "A heuristic failure does not prove that no feasible physical arrangement exists."
```


test cases.md - # **Automated Test Cases**

Automated tests are in `inventory/tests.py`.

| ID | Scenario | Expected behavior |

TC-01	Single product fits the cheapest box	Recommend the cheapest qualifying box
TC-02	Product requires axis-aligned rotation	Packing succeeds if the rotated dimensions fit
TC-03	Product is too large for candidate box	Candidate is rejected
TC-04	Order exceeds a box's weight limit	Box is rejected and a later candidate may be selected
TC-05	Multiple items fit without overlap	Return a placement for every item
TC-06	No available box can fit item	Return `found: false`
TC-07	Empty order	Return HTTP 400
TC-08	Unknown product ID	Return HTTP 400
TC-09	Duplicate product entries	Return HTTP 400 and request one entry with quantity
TC-10	Quantity is zero	Return HTTP 400
TC-11	Product and box list endpoints	Return HTTP 200

Run with `python manage.py test`. Add additional tests after you manually review edge cases.

Manual API Test Cases

1. Valid Order

- Input: One smartphone and two headphones.
- Expected: Recommend the lowest-cost box the packing heuristic can fit all items into.
- Actual: Medium Box recommended at ₹50.
- Status: Passed.

2. Zero Quantity

- Input: Product ID 1 with quantity 0.
- Expected: HTTP 400 validation error.
- Actual: HTTP 400; quantity must be at least 1.
- Status: Passed.

3. Unknown Product ID

- Input: Product ID 10, which does not exist.
- Expected: HTTP 400 validation error.
- Actual: HTTP 400; unknown product ID reported.
- Status: Passed.

4. Oversized Product

- Input: Oversized Monitor (100 × 80 × 60 cm).
- Expected: No box recommendation if the packing heuristic cannot fit the product.
- Actual: HTTP 200; `found: false`.
- Status: Passed, with heuristic limitation noted.

5. Order Exceeds Weight Capacity

- Input: Heavy Test Item weighing 26 kg.
- Expected: No box recommendation because every box has a maximum weight below 26 kg.
- Actual: HTTP 200; `found: false`. All five boxes were rejected for exceeding weight capacity.
- Status: Passed.

I think now it's all done so can we move towards the next step

ChatGPT said:

You said:

Test Run Output

Environment

- OS: Windows 11
- Python version: Python 3.11.6
- Django version: 5.2.18

Automated Test Results

Command: `python manage.py test`

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py test
Found 11 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
.....C:\Users\Shan\Downloads\box_selection_system_starter\box_sele
ction_system\.venv\Lib\site-packages\rest_framework\fields.py:992:
UserWarning: min_value should be an integer or Decimal instance.
  warnings.warn("min_value should be an integer or Decimal
instance.")
......
----------------------------------------------------------------------
Ran 11 tests in 0.034s

OK
Destroying test database for alias 'default'...
```

Django System Check

Command: `python manage.py check`

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py check
System check identified no issues (0 silenced).
```

Manual API Test Results

`Summarization` : In below are the test cases, that has been tested by giving the correct valid order values, quantity as 0, unknown pr0duct id that is not mentioned or added in the product (product_id = 15), product that is too large for every box to carry by, the order exceeds every boxes weight limits here by these are the test cases conducted on this project.

Actual terminal output:

1. Valid Order

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
"found": true,
"box": {
    "id": 2,
    "name": "Medium Box",
    "cost_inr": "50.00",
    "max_weight_kg": "5.000"
},
"total_order_weight_kg": "1.200",
"packing_method": "3d_corner_point_heuristic",
"placements": [
    {
        "item_key": "3-1",
        "position_cm": {
            "x": 0.0,
            "y": 0.0,
            "z": 0.0
        },
        "oriented_dimensions_cm": {
            "length": 8.0,
            "width": 18.0,
            "height": 20.0
        }
    },
    {
        "item_key": "3-2",
        "position_cm": {
            "x": 8.0,
            "y": 0.0,
            "z": 0.0
        },
        "oriented_dimensions_cm": {
            "length": 8.0,
            "width": 18.0,
            "height": 20.0
        }
```

```
        },
        {
            "item_key": "1-1",
            "position_cm": {
                "x": 16.0,
                "y": 0.0,
                "z": 0.0
            },
            "oriented_dimensions_cm": {
                "length": 2.0,
                "width": 8.0,
                "height": 15.0
            }
        }
    ],
    "note": "Lowest-cost box accepted by the packing heuristic; not a
proof of globally optimal packing."
```


2. QUANTITY = 0

HTTP 400 Bad Request
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "error": "Invalid request",
    "details": {
        "items": [
            {
                "quantity": [
                    "Ensure this value is greater than or equal to 1."
                ]
            }
        ]
    }
}
```


3. UNKNOWN PRODUCT ID = 15

HTTP 400 Bad Request
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "error": "Invalid request",
    "details": {
        "items": [
            "Unknown product_id value(s): [15]"
        ]
    }
}
```


4. PRODUCT TOO LARGE FOR EVERY BOX - OVERSIZED MONITOR

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "found": false,
    "message": "No available box passed the weight check and packing heuristic.",
    "total_order_weight_kg": "3.000",
    "rejected_boxes": [
        {
            "box": "Small Box",
            "reason": "order_exceeds_weight_capacity"
        },
        {
            "box": "Medium Box",
            "reason": "heuristic_could_not_pack_all_items"
        },
        {
            "box": "Large Box",
            "reason": "heuristic_could_not_pack_all_items"
        },
        {
            "box": "Heavy Duty Box",
```

```
            "reason": "heuristic_could_not_pack_all_items"
        },
        {
            "box": "Extra Large Box",
            "reason": "heuristic_could_not_pack_all_items"
        }
    ],
    "packing_method": "3d_corner_point_heuristic",
    "limitation": "A heuristic failure does not prove that no feasible physical arrangement exists.
}
```


5. Additional validation check - QUANTITY - 101

HTTP 400 Bad Request
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept


```
{
    "error": "Invalid request",
    "details": {
        "items": [
            {
                "quantity": [
                    "Ensure this value is less than or equal to 100."
                ]
            }
        ]
    }
}
```


Weight - limit test

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept

"found": false,
"message": "No available box passed the weight check and packing heuristic.",
"total_order_weight_kg": "26.000",
"rejected_boxes": [
{
"box": "Small Box",
"reason": "order_exceeds_weight_capacity"
},
{
"box": "Medium Box",
"reason": "order_exceeds_weight_capacity"
},
{
"box": "Large Box",
"reason": "order_exceeds_weight_capacity"
},
{
"box": "Heavy Duty Box",
"reason": "order_exceeds_weight_capacity"
},
{
"box": "Extra Large Box",
"reason": "order_exceeds_weight_capacity"
}
],
"packing_method": "3d_corner_point_heuristic",
"limitation": "A heuristic failure does not prove that no feasible physical arrangement exists."

test_cases.md - # **Automated Test Cases**

Automated tests are in `inventory/tests.py`.

| ID | Scenario | Expected behavior |

| TC-01 | Single product fits the cheapest box | Recommend the cheapest qualifying box |
| TC-02 | Product requires axis-aligned rotation | Packing succeeds if the rotated dimensions fit |
| TC-03 | Product is too large for candidate box | Candidate is

rejected |
TC-04	Order exceeds a box's weight limit	Box is rejected and a later candidate may be selected
TC-05	Multiple items fit without overlap	Return a placement for every item
TC-06	No available box can fit item	Return `found: false`
TC-07	Empty order	Return HTTP 400
TC-08	Unknown product ID	Return HTTP 400
TC-09	Duplicate product entries	Return HTTP 400 and request one entry with quantity
TC-10	Quantity is zero	Return HTTP 400
TC-11	Product and box list endpoints	Return HTTP 200

Run with `python manage.py test`. Add additional tests after you manually review edge cases.

Manual API Test Cases

1. Valid Order

- Input: One smartphone and two headphones.
- Expected: Recommend the lowest-cost box the packing heuristic can fit all items into.
- Actual: Medium Box recommended at ₹50.
- Status: Passed.

2. Zero Quantity

- Input: Product ID 1 with quantity 0.
- Expected: HTTP 400 validation error.
- Actual: HTTP 400; quantity must be at least 1.
- Status: Passed.

3. Unknown Product ID

- Input: Product ID 15, which does not exist.
- Expected: HTTP 400 validation error.
- Actual: HTTP 400; unknown product ID reported.
- Status: Passed.

4. Oversized Product

- Input: Oversized Monitor (100 × 80 × 60 cm).
- Expected: No box recommendation if the packing heuristic cannot fit the product.
- Actual: HTTP 200; `found: false`.
- Status: Passed, with heuristic limitation noted.

5. Order Exceeds Weight Capacity

- Input: Heavy Test Item weighing 26 kg.
- Expected: No box recommendation because every box has a maximum weight below 26 kg.
- Actual: HTTP 200; `found: false`. All five boxes were rejected for exceeding weight capacity.
- Status: Passed.
  I guess it's all done let's move on to step 2

ChatGPT said:

You said:

```
from rest_framework import serializers
from .models import Product, ShippingBox


class ProductSerializer(serializers.ModelSerializer):
    class Meta:
        model = Product
        fields = ["id", "name", "length_cm", "width_cm", "height_cm",
"weight_kg"]


class ShippingBoxSerializer(serializers.ModelSerializer):
    class Meta:
        model = ShippingBox
        fields = [
            "id", "name", "inner_length_cm", "inner_width_cm",
            "inner_height_cm", "max_weight_kg", "cost_inr",
        ]


class OrderItemSerializer(serializers.Serializer):
```

```
    product_id = serializers.IntegerField(min_value=1)
    quantity = serializers.IntegerField(min_value=1, max_value=100)


class RecommendationRequestSerializer(serializers.Serializer):
    items = OrderItemSerializer(many=True, allow_empty=False)

    def validate_items(self, items):
        ids = [item["product_id"] for item in items]
        if len(ids) != len(set(ids)):
            raise serializers.ValidationError(
                "Use one entry per product and set its quantity."
            )
        products = Product.objects.in_bulk(ids)
        missing = sorted(set(ids) - set(products))
        if missing:
            raise serializers.ValidationError(
                f"Unknown product_id value(s): {missing}"
            )
        return items
```

ChatGPT said:

You said:

```
class OrderItemSerializer(serializers.Serializer):
    product_id = serializers.IntegerField(min_value=1)
    quantity = serializers.IntegerField(min_value=1, max_value=100)
```

and in the test_output.md - min_value should be an integer or Decimal instance.

```
  warnings.warn("min_value should be an integer or Decimal instance.")
```

this is the result i got

ChatGPT said:

You said:

```
from django.core.validators import MinValueValidator
from django.db import models


class Product(models.Model):
    name = models.CharField(max_length=120, unique=True)
    length_cm = models.DecimalField(max_digits=8,
```

```
decimal_places=2, validators=[MinValueValidator(0.01)])
    width_cm = models.DecimalField(max_digits=8,
decimal_places=2, validators=[MinValueValidator(0.01)])
    height_cm = models.DecimalField(max_digits=8,
decimal_places=2, validators=[MinValueValidator(0.01)])
    weight_kg = models.DecimalField(max_digits=8,
decimal_places=3, validators=[MinValueValidator(0.001)])

    class Meta:
        ordering = ["name"]

    def __str__(self):
        return self.name


class ShippingBox(models.Model):
    name = models.CharField(max_length=120, unique=True)
    inner_length_cm = models.DecimalField(max_digits=8,
decimal_places=2, validators=[MinValueValidator(0.01)])
    inner_width_cm = models.DecimalField(max_digits=8,
decimal_places=2, validators=[MinValueValidator(0.01)])
    inner_height_cm = models.DecimalField(max_digits=8,
decimal_places=2, validators=[MinValueValidator(0.01)])
    max_weight_kg = models.DecimalField(max_digits=8,
decimal_places=3, validators=[MinValueValidator(0.001)])
    cost_inr = models.DecimalField(max_digits=10,
decimal_places=2, validators=[MinValueValidator(0)])

    class Meta:
        ordering = ["cost_inr", "name"]

    def __str__(self):
        return f"{self.name} (₹{self.cost_inr})"
```

ChatGPT said:

You said:

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py test
Found 11 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
.....C:\Users\Shan\Downloads\box_selection_system_starter\box_sele
```

```
ction_system\.venv\Lib\site-packages\rest_framework\fields.py:992:
UserWarning: min_value should be an integer or Decimal instance.
  warnings.warn("min_value should be an integer or Decimal
instance.")
......
----------------------------------------------------------------------
Ran 11 tests in 0.045s

OK
Destroying test database for alias 'default'...

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py check
System check identified no issues (0 silenced).

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>
```

ChatGPT said:

You said:

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python -W error::UserWarning manage.py test
Found 11 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
.....E.....
======================================================
======================================
ERROR: test_product_and_box_list_endpoints
(inventory.tests.RecommendationAPITests.test_product_and_box_list
_endpoints)
----------------------------------------------------------------------
Traceback (most recent call last):
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\inventory\tests.py", line 120, in
test_product_and_box_list_endpoints
    self.assertEqual(self.client.get("/api/products/").status_code, 200)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File
```

```
"C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\test.py", line 292, in get
    response = super().get(path, data=data, **extra)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\test.py", line 209, in get
    return self.generic('GET', path, **r)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\test.py", line 237, in generic
    return super().generic(
           ^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\django\test\client.py", line 671, in generic
    return self.request(**r)
           ^^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\test.py", line 289, in request
    return super().request(**kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\test.py", line 241, in request
    request = super().request(**kwargs)
              ^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\django\test\client.py", line 1090, in request
    self.check_exception(response)
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\django\test\client.py", line 805, in check_exception
    raise exc_value
```

```
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-
packages\django\core\handlers\exception.py", line 55, in inner
    response = get_response(request)
               ^^^^^^^^^^^^^^^^^^^^^
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\django\core\handlers\base.py",
line 197, in _get_response
    response = wrapped_callback(request, *callback_args,
**callback_kwargs)

^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^^^^^^^^^^^^^^^^
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-
packages\django\views\decorators\csrf.py", line 65, in
_view_wrapper
    return view_func(request, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\django\views\generic\base.py",
line 105, in view
    return self.dispatch(request, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\rest_framework\views.py", line
515, in dispatch
    response = self.handle_exception(exc)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\rest_framework\views.py", line
475, in handle_exception
    self.raise_uncaught_exception(exc)
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\rest_framework\views.py", line
486, in raise_uncaught_exception
    raise exc
  File
```

```
"C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\views.py", line 512, in dispatch
    response = handler(request, *args, **kwargs)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\generics.py", line 203, in get
    return self.list(request, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\mixins.py", line 46, in list
    return Response(serializer.data)
                    ^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\serializers.py", line 797, in data
    ret = super().data
          ^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\serializers.py", line 251, in data
    self._data = self.to_representation(self.instance)

^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\serializers.py", line 715, in to_representation
    return [
           ^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\serializers.py", line 716, in <listcomp>
    self.child.to_representation(item) for item in iterable
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system\.venv\Lib\site-packages\rest_framework\serializers.py",
```

```
line 525, in to_representation
    for field in fields:
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\rest_framework\serializers.py",
line 386, in _readable_fields
    for field in self.fields.values():
                 ^^^^^^^^^^^^
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\django\utils\functional.py", line
47, in __get__
    res = instance.__dict__[self.name] = self.func(instance)
                                         ^^^^^^^^^^^^^^^^^^^
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\rest_framework\serializers.py",
line 374, in fields
    for key, value in self.get_fields().items():
                      ^^^^^^^^^^^^^^^^^
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\rest_framework\serializers.py",
line 1115, in get_fields
    fields[field_name] = field_class(**field_kwargs)
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File
"C:\Users\Shan\Downloads\box_selection_system_starter\box_select
ion_system\.venv\Lib\site-packages\rest_framework\fields.py", line
992, in __init__
    warnings.warn("min_value should be an integer or Decimal
instance.")
UserWarning: min_value should be an integer or Decimal instance.

----------------------------------------------------------------------
Ran 11 tests in 0.056s

FAILED (errors=1)
Destroying test database for alias 'default'...

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>
```

ChatGPT said:

You said:

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py test
Found 11 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
.....C:\Users\Shan\Downloads\box_selection_system_starter\box_sele
ction_system\.venv\Lib\site-packages\rest_framework\fields.py:992:
UserWarning: min_value should be an integer or Decimal instance.
  warnings.warn("min_value should be an integer or Decimal
instance.")
.....
----------------------------------------------------------------------
Ran 11 tests in 0.050s

OK
Destroying test database for alias 'default'...    code - from decimal
import Decimal
from rest_framework import serializers
from .models import Product, ShippingBox


class ProductSerializer(serializers.ModelSerializer):
    length_cm = serializers.DecimalField(
        max_digits=8, decimal_places=2, min_value=Decimal("0.01")
    )
    width_cm = serializers.DecimalField(
        max_digits=8, decimal_places=2, min_value=Decimal("0.01")
    )
    height_cm = serializers.DecimalField(
        max_digits=8, decimal_places=2, min_value=Decimal("0.01")
    )
    weight_kg = serializers.DecimalField(
        max_digits=8, decimal_places=3, min_value=Decimal("0.001")
    )

    class Meta:
        model = Product
        fields = ["id", "name", "length_cm", "width_cm", "height_cm",
"weight_kg"]
```

```
class ShippingBoxSerializer(serializers.ModelSerializer):
    class Meta:
        model = ShippingBox
        fields = [
            "id", "name", "inner_length_cm", "inner_width_cm",
            "inner_height_cm", "max_weight_kg", "cost_inr",
        ]


class OrderItemSerializer(serializers.Serializer):
    product_id = serializers.IntegerField(min_value=1)
    quantity = serializers.IntegerField(min_value=1, max_value=100)


class RecommendationRequestSerializer(serializers.Serializer):
    items = OrderItemSerializer(many=True, allow_empty=False)

    def validate_items(self, items):
        ids = [item["product_id"] for item in items]
        if len(ids) != len(set(ids)):
            raise serializers.ValidationError(
                "Use one entry per product and set its quantity."
            )
        products = Product.objects.in_bulk(ids)
        missing = sorted(set(ids) - set(products))
        if missing:
            raise serializers.ValidationError(
                f"Unknown product_id value(s): {missing}"
            )
        return items
```

ChatGPT said:

You said:

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>python manage.py test
Found 11 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
...........
----------------------------------------------------------------------
```

```
Ran 11 tests in 0.042s

OK
Destroying test database for alias 'default'...

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>
```

ChatGPT said:

You said:

```
(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system>dir
 Volume in drive C is OS
 Volume Serial Number is 9A07-ADBA

 Directory of
C:\Users\Shan\Downloads\box_selection_system_starter\box_selecti
on_system

19-10-2026  19:09    <DIR>          .
19-10-2026  18:08    <DIR>          ..
19-10-2026  18:08    <DIR>          .github
19-10-2026  18:08                84 .gitignore
19-10-2026  18:13    <DIR>          .venv
19-10-2026  18:08             2,532 AI_USAGE.md
19-10-2026  18:15    <DIR>          config
19-10-2026  19:09           147,456 db.sqlite3
19-10-2026  18:26    <DIR>          inventory
19-10-2026  18:08               247 manage.py
19-10-2026  18:08             4,004 README.md
19-10-2026  18:08                49 requirements.txt
19-10-2026  18:08               989 SUBMISSION_NOTES.md
19-10-2026  20:34             2,104 TEST_CASES.md
19-10-2026  20:36             5,836 TEST_OUTPUT.md
19-10-2026  18:08               373 WHAT_I_LEARNED_TEMPLATE.txt
              10 File(s)        163,674 bytes
               6 Dir(s)  304,536,461,312 bytes free

(.venv)
```

C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>

ChatGPT said:

You said:

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>git status
fatal: not a git repository (or any of the parent directories): .git

ChatGPT said:

You said:

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>git status
On branch master
nothing to commit, working tree clean

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>

ChatGPT said:

You said:

github doesn't updated the project yet i think

ChatGPT said:

You said:

My git remote -v output is empty. Guide me through connecting my project to GitHub.

ChatGPT said:

You said:

everything is updated in the github so what's next we are going to do

ChatGPT said:

You said:

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>python manage.py check
System check identified no issues (0 silenced).

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>python manage.py test
Found 11 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
....

Ran 11 tests in 0.048s

OK
Destroying test database for alias 'default'...

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified: TEST_OUTPUT.md

no changes added to commit (use "git add" and/or "git commit -a")

(.venv)
C:\Users\Shan\Downloads\box_selection_system_starter\box_selection_system> and step 2 if fully checked and everything mentioned in here is there

ChatGPT said: